PALEOMAGNETISM OF THE WESTCOAST COMPLEX AND THE GEOTECTONICS OF THE VANCOUVER ISLAND SEGMENT OF THE WRANGELLIAN SUBTERRANE

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ABSTRACT

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Paleomagnetic data from 46 sites (674 specimens) of the Westcoast Crystalline Gneiss Complex on the west coast of Vancouver Island using AF and thermal demagnetization methods yields a high blocking temperature WCB component ($>560°C$) with a pole at 335°W, 66°N ($\delta_p = 4°$, $\delta_m = 6°$) and a lower coercivity WCA component (~ 25 mT, $<500°C$) with a pole at 52°W, 79°N ($\delta_p = 7°$, $\delta_m = 8°$). Further thermal demagnetization data from 24 sites in the Jurassic Island Intrusions also defines two high blocking temperature components. The IIA component pole is at 59°W, 79°N ($\delta_p = 7°$, $\delta_m = 8°$) and IIB pole at 130°W, 73°N ($\delta_p = 12°$, $\delta_m = 13°$). Combined with previous data from the Karmutsen Basalts and mid-Tertiary units on Vancouver Island and from the adjacent Coast Plutonic Complex, the geotectonic motions are examined for the Vancouver Island segment of the Wrangellian Subterrane of composite Terrane II of the Cordillera. The simplest hypothesis invokes relatively uniform translation for Terrane II from Upper Triassic to Eocene time producing $\sim 39° \pm 6°$ of northward motion relative to the North American craton, combined with $\sim 40°$ of clockwise rotation during the Lower Tertiary.

INTRODUCTION

The Westcoast Crystalline Complex crops out along the west coast of Vancouver Island (Fig. 1) and forms part of the Wrangellia Subterrane of composite Terrane II (Monger *et al.* 1982) of the Canadian Cordillera. Paleomagnetic studies of the Triassic Karmutsen Basalts indicate that Terrane II has undergone $\sim 40°$ of northward latitudinal displacement since the time of extrusion. Noting that the Westcoast Complex yields a late Paleozoic U-Pb zircon age of 264 Ma, this paleomagnetic study was undertaken in an attempt to further extend the record of the Terrane II geotectonic motion.

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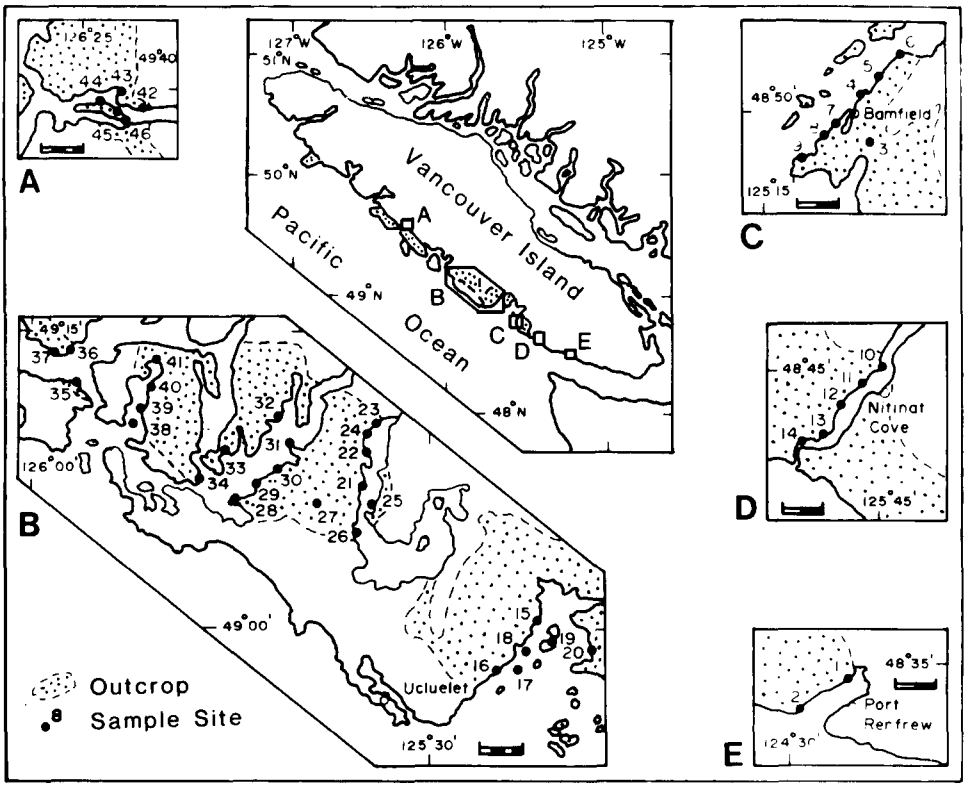


Fig. 1. Vancouver Island (top center) giving the locations of the inset maps (A to E) which show the sampling sites in the Westcoast Complex (stipple). The scale bars are in kilometers.

GEOLOGY

The geology of Vancouver Island has recently been summarized by Muller (1977). Siluro-Devonian migmatites with U-Pb ages of 390 to 450 Ma are exposed on its southeastern shore in the Tyee, Wark, Colquitz and Turtleback complexes. They are part of the adjacent Alexander Subterrane which formed as a volcanic-plutonic arc far distant (2000 km?) from the inboard Cordillera. After the Caribooan-Antler Orogeny, late Paleozoic eugeosynclinal conglomerates, basalts and shelf sediments were deposited outboard of the Alexander subterrane. During the Permo-Triassic Tahtanian-Sonoma Orageny these rocks were metamorphosed to form the Westcoast Complex. This basic heterogenous assemblage gives a U-Pb zircon age of 264 ± 7 Ma (Muller *et al.* 1974).

During Middle to Upper Triassic rifting or ~ 230 Ma ago (Odin 1982),

~600 m of Karmutsen tholeiitic plateau basalts were extruded over Vancouver Island. After Upper Triassic deposition of thin clastic carbonate sediments, the Lower Jurassic Bonanza volcanics were extruded which give a Rb-Sr isochron age of 188 ± 5 Ma (Muller 1977) (ages recalculated using constants recommended by Steiger and Jäger (1977)). Coincidently in early Jurassic time, the Westcoast Complex was migmatized at deep crustal levels - possibly with further intrusion - to yield massive to foliated hornblende-plagioclase gneisses, agmatites and amphibolites with numerous zenoliths and minor late veins and dikes. One late amphibolite dike gives a minimum K-Ar age of 196 Ma. This migmatization event may also have generated the granitoid magma which intruded at higher crustal levels to form the extensive Island Intrusion batholiths which give a Lower Jurassic Rb-Sr isochron age of 178 ± 10 Ma (R. L. Armstrong & K. Scott, pers. comm. in Muller 1977). K-Ar dates indicate significant subsequent thermal events at ~155, ~110 and ~40 Ma with related intrusions and/or metamorphism.

METHODS

Five or six cores, oriented by sun compass, were collected at each of 46 sites representing many rock types in the Complex (Fig. 1, Table 1). Generally three specimens were sliced from each core. After measuring the natural remanent magnetization (NRM) of all 674 specimens, one representative specimen per site was alternating field (AF) demagnetized in 15 steps to 100 mT and a second specimen was thermally demagnetized in 10 steps to 600°C. All remaining specimens were first AF cleaned at an assigned field between 5 and 40 mT and again thermally cleaned at 560°C using a Schonstedt SSM-1A spinner magnetometer, a GSD-1 AF demagnetizer and a TSD-1 thermal demagnetizer.

AF DEMAGNETIZATION

Population level screening (Kamb 1959, Van Alstine 1980, Stupavsky and Symons 1982) using a 1% smoothing circle to contour the NRM directions gives a "bull's eye" anomaly (>12%, D = 40°, I = 70°; peak, declination, inclination) (Fig. 2a). This anomaly is close to the present Earth's magnetic field (EMF) direction of 23.1°, 70.5° which suggests that viscous remanence (VRM) components are present. Most NRM intensities are in the 10^{-1} to 1 Am^{-1} range.

The low median destructive field of ~7 mT on AF step demagnetization (Fig. 3a) further indicates a significant VRM contribution to the NRM, however, the mean intensity ratio decay curve flattens out and remains at a

TABLE I

Summary of lithology and tiered statistics by sites for AF data from the Westcoast Complex

Site	Lithology	AF	N^1,N	Mean Remanence Decl. o	Incl. o	α_{95} o	k	Site	Lithology	AF	N^1,N	Mean Remanence Decl. o	Incl. o	α_{95} o	k
1	mt bv	25	5,1	282	58	–	–	24	mt di	10	4,1	95	77	–	–
2	mt bv	5	5,3	111	77	13	94	25	gd,gr	10	4,4	39	72	9	100
3	fo gd-di	15	3,3	34	81	15	70	26	fo gb	30	5,3	345	80	47	8
4	fo gd-di	15	5,5	17	65	13	36	27	gb	30	5,5	193	10	9	78
5	fo di	25	3,3	358	72	18	47	28	gn di	25	5,5	125	26	9	77
6	mt iv	30	5,2	112	11	21	147	29	fo gb	30	4,3	93	44	35	14
7	fo di	25	4,3	46	50	17	55	30	db,di	10	4,3	29	44	31	17
8	fo di	30	5,5	236	–31	21	14	31	fo di	30	4,4	19	64	17	31
9	fo di	40	5,3	5	15	16	58	32	gd	30	4,3	11	47	21	37
10	gb	30	4,3	289	–52	10	150	33	gn di	5	4,4	72	67	23	16
11	gb	25	5,5	294	–26	19	17	34	gn di	10	4,3	3	64	10	151
12	gb, gr	15	4,3	13	71	13	86	35	gn di	50	4,3	175	–45	55	6
13	mt gb	10	5,5	22	56	24	11	36	gn di	5	4,2	160	–56	27	5

14	mt iv	5	5,2	3	55	52	25	37	di	10	4,4	57	47	32	9
15	gb,di	25	5,5	13	42	7	121	38	mt bv	5	4,0	–	–	–	–
16	gd, db	10	5,4	338	66	14	42	39	gb	25	4,3	76	4	14	84
17	fo gd	15	5,2	1	74	49	27	40	fo di	25	4,3	6	74	30	18
18	di, db	30	5,3	18	10	25	25	41	gn di	10	4,4	163	3	88	2
19	mt di	25	5,4	35	16	6	266	42	di, gd	15	4,4	33	76	14	44
20	gd, di	20	5,5	17	68	9	76	43	di, gd	25	4,4	5	64	11	70
21	gd	40	5,5	11	64	16	25	44	gd	25	4,4	22	56	5	321
22	gd	20	5,4	21	69	14	44	45	gd	25	4,4	9	63	9	112
23	mt bv	30	5,5	24	71	5	290	46	gd	20	4,4	2	64	9	98
									MEAN	(23)		19.2	66.4	5.0	37.8

Notes: Lithologic abbreviations: bv – basic volcanics, di –diorite, fo – foliated, gb – gabbro, gd – granodiorite, gn – gneissic, gr – granite, iv – intermediate volcanics, mt – metamorphosed.

The alternating field (AF) is H in millitesla. N,N are the number of cores collected and core means averaged respectively. The mean remanence direction is given by its declination (Decl.), inclination (Incl.), radius of cone 95% confidence (α_{95}) in degrees (°) and precision parameter (k) (Fisher 1953). The Complex MEAN remanence direction utilizes (23) means from sites, 2, 3, 4, 5, 7, 12, 13, 15, 16, 20, 21, 22, 23, 25, 31, 32, 33, 34, 42, 43, 44, 45 and 46.

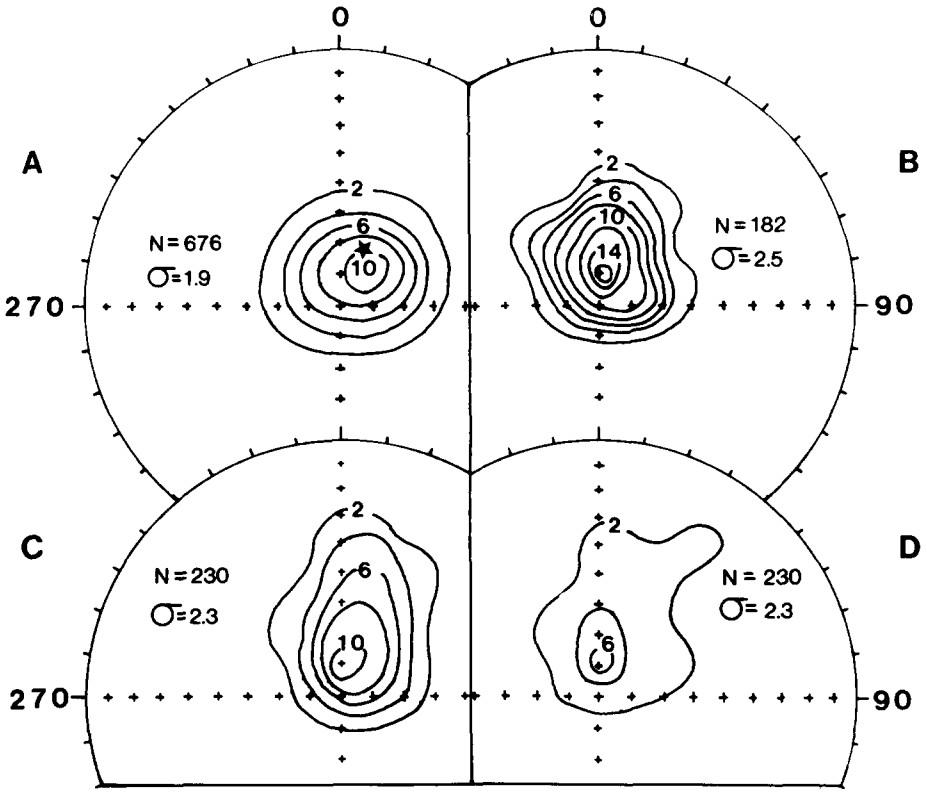


Fig. 2. Stereocontour plots on the lower hemisphere of Schmidt equal area stereonets using a 1% smoothing area for the: a) NRM directions; and the AF step demagnetization directions between b) 2.5 and 10 mT, c) 15 and 40 mT, and d) 50 and 100 mT. N is the number of vectors plotted. δ is the $E + 2\delta$ or 95% confidence contour. The star locates the present geomagntic field direction. There are no statistically significant concentrations in the upper hemisphere.

relatively high 0.14 ± 0.19 (s.d.; standard deviation) at 100 mT. The large standard deviation limits reflect the wide range of rock types. After changing the polarity of seven reversed sites, the mean remanence direction (Fig. 3b) shallows to a consistent $D = \sim 14°$, $I = 59°$ through the 20 to 70 mT range. Above ~ 50 mT, the cone of 95% confidence (α_{95}) increases to $> 13°$ and precision parameter (k, Fisher 1953) decreases to < 2.3 thereby indicating scattering as anhysteretic remanence (ARM) components are acquired in the AF demagnetizing process. When the 2.5 to 10, 15 to 40, and 50 to 100 mT residual directions are grouped (Fig. 2b, c, d), the anomaly mode or A component at $D = \sim 8°$, $I = 76°$ is progressively diminished in magnitude as it is removed. The elongation of the mode towards $\sim 10°$ declination suggests that a second underlying or B component is present. The vectors removed (VR) between successive AF steps were then plotted in 0 to 15, 15

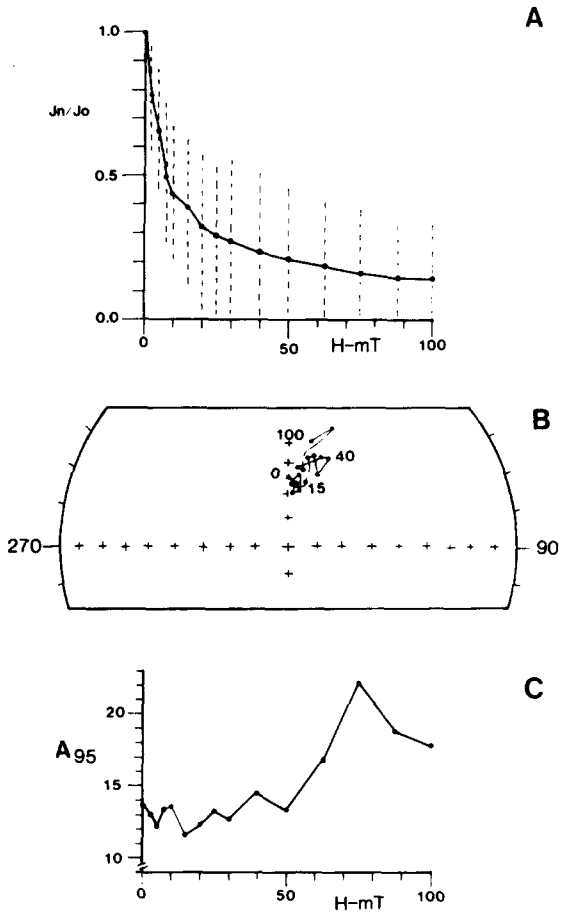


Fig. 3. Mean AF step demagnetization data for 46 pilot specimens showing: a) the mean ratio of the measured to original remanence intensity (J_n/J_o) with its standard deviation bar as a function of the demagnetizing field (H-mT); b) the mean remanence direction at each step from 0 to 100 mT with conventions as in Figure 2; and c) the radius of the cone of 95% confidence (α_{95}, Fisher 1953) for each step.

to 40, and 40 to 100 mT groups (Fig. 4a, b, c). Up to 40 mT only the steeply inclined A component is being removed, but above 40 mT the B component at $D = \sim 14°$, $I = 47°$ is also being removed.

AF cleaning of all specimens in the 15 to 40 mT range gives a well defined A component mode ($>12\%$ $D = 18°$, $I = 69°$) (Fig. 4d) of $>>99\%$ confidence which is isolated using conventional tiered statistics on sites for which $\alpha_{95} < 25°$ (Fig. 5a, Table 1). Specimens from five sites have specimens with both A and B components.

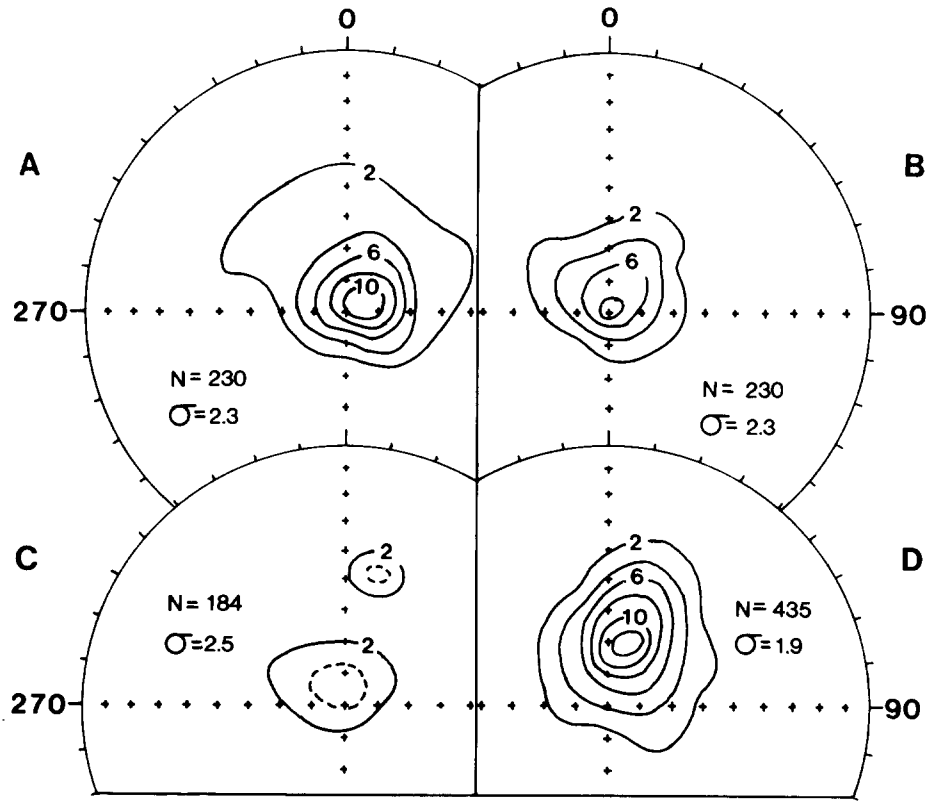


Fig. 4. Stereocontour plots of the directions of the vectors removed between AF demagnetization steps from: a) 0 to 15 mT; b) 15 to 40 mT; and c) 40 to 100 mT; and d) of the directions after "one step" AF cleaning at the optimum field in the 5 to 40 mT range. Conventions as in Figure 2. The dashed contour in 4c is the 3% density contour.

THERMAL DEMAGNETIZATION

The mean intensity ratio decay curve on thermal step demagnetization (Fig. 6) shows a rapid drop up to ~400°C as VRM is removed, a plateau at ~0.18 from 400 to 560°C as stable magnetite remanence carriers are isolated, and a drop to ~0.06 at 600°C. The VR directions from 200 to 500°C steps (Fig. 5b) give a well defined A component mode (>7%, D = 31°, I = 7°) while the residual vectors for the same temperature range (Fig. 7a) give an elongate hybrid A plus B component anomaly. However, the VR and residual directions (Fig. 7b, c) for the 500 to 575°C show the separation and removal of both A and B components with the B component at D = 17°, I = 47°. Thermal cleaning of all specimens at 560°C gives a well defined circular B component mode (>6%, D = 15°, I = 47°). The minor

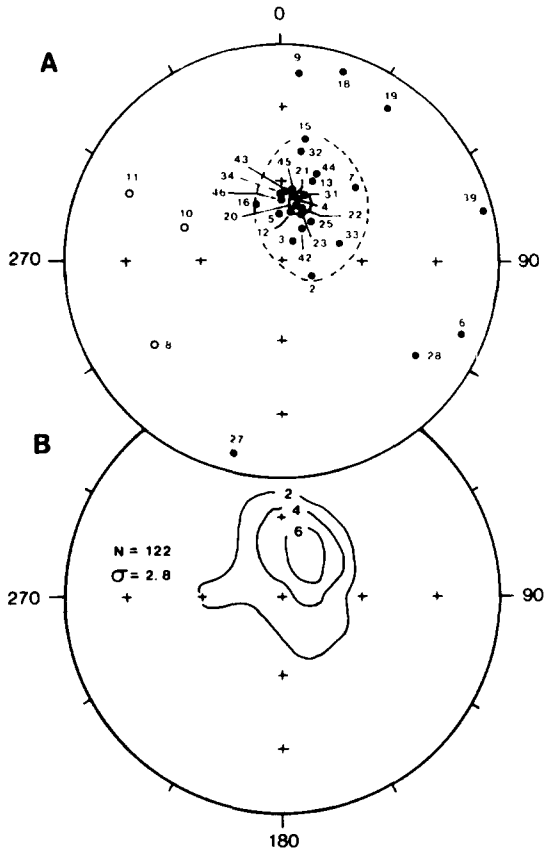


Fig. 5. a) Site mean remanence directions calculated using conventional tiered statistics with the mean direction of those site means falling within the dash line averaged to give the unit mean of D = 19.2°, I = 66.4° ($\alpha_{95} = 5.0°$) shown by a triangle with its cone of 95% confidence; and b) thermal step demagnetization data showing the stereocountour plot of the vectors removed between steps in the 200 to 500°C range. Conventions as in Figures 2 and 3.

lobe in the 2% countour towards steeper inclinations indicates that a very minor fraction of the A component resides in domains with blocking temperatures approaching 560°C.

DISCUSSION

Westcoast Complex Poles

The *W*estcoast A component (WCA) pole using the 5 to 40 mT AF cleaning data is located at 52.0°W, 79.3°N ($\delta_p = 6.7°$, $\delta_m = 8.0°$) (Table 2) and resides dominantly in domains with blocking temperatures up to ~500°C.

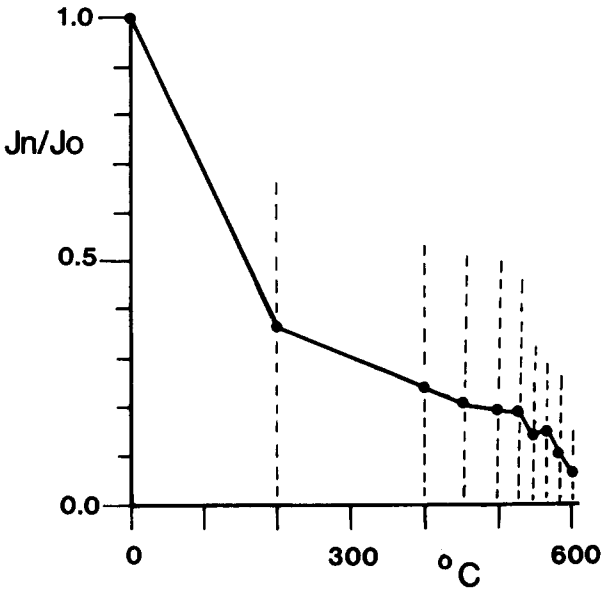


Fig. 6. Thermal step demagnetization showing the mean J_n/J_o ratio at each step (T-°C). Conventions as in Figure 3.

The WCB pole using the 560°C thermal cleaning data is located at 335.2°W, 66.1°N ($\delta_p = 4.1°$, $\delta_m = 6.4°$) and is clearly the stable high temperature magnetite component. Given the large number and widespread distribution of the sites, the tilt corrections are assumed to be self-cancelling.

Karmutsen Basalt Poles

Paleomagnetic studies on the Middle to Upper Triassic Karmutsen Basalts (Carlisle and Susuki 1974) by Symons (1971a), Irving and Yole (1972), Schwarz *et al.* (1980) and Yole and Irving (1980) all isolate two components. The A component is steeply inclined to the north, resides in lower coercivity domains, and is deemed to be metamorphic in origin. Its four poles (KB1A, KB2A, KB3A and KB4A in Fig. 8a) give a site weighted mean pole (KBA) at 8°W, 81°N ($\alpha_{95} = 6°$) which reflects a metamorphic thermal event. The B component is shallowly inclined to the north, resides in high coercivity domains, and is deemed to be the primary Triassic remanence. Its poles (KB1B, KB2B, KB3Ba, KB3Bb, KB3Bc, KB4Ba and KB4Bb), excluding KB3Bb found at one site, define a small circle arc of $108° \pm 6°$ from Vancouver Island. This arc is $\sim 39° \pm 6°$ far-sided with respect to the 230 Ma reference pole for cratonic North America (Irving and

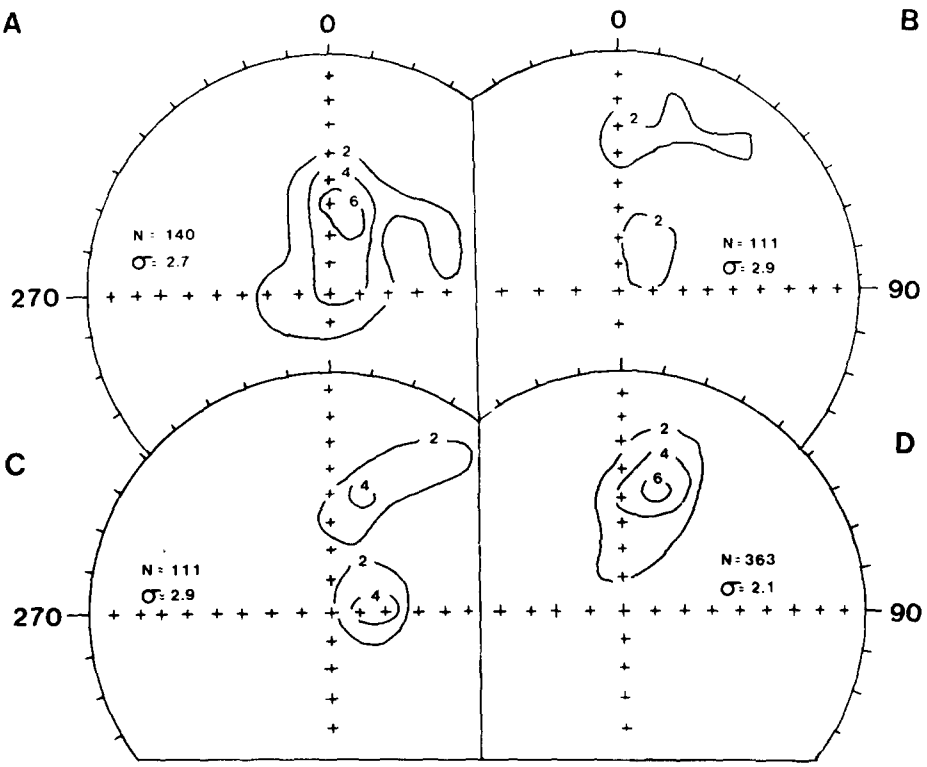


Fig. 7. Stereocontour plots for thermal step demagnetization data showing the: a) measured directions for the 200 to 500°C steps; b) the vectors removed between steps in the 500 to 575°C range; c) measured directions for the 525 to 575°C steps; and d) the measured directions for "one step" thermal cleaning at 560°C. Conventions as in Figure 2.

TABLE II

Component mean directions and poles

Component	Treatment	Number of Sites, Specimens	Mean Remanence Direction				Pole Position			
			R	Decl. o	Incl. o	α_{95} o	Long. o_W	Lat. o_N	δ_p o	δ_m o
WCA	AF, 5-40 mT	39, 233	37.32	16.7	58.2	4.7	52.0	79.3	6.7	8.0
WCB	Ther. 560°C	27, 69	26.11	12.8	46.6	5.0	335.2	66.1	4.1	6.4
IIA	Ther. 450°C	11, 22	10.79	17.6	69.5	4.4	59.	79.	7.	8.
IIB	Ther. 450°C	10, 24	9.61	7.8	50.1	10.2	325.	71.	9.	14.

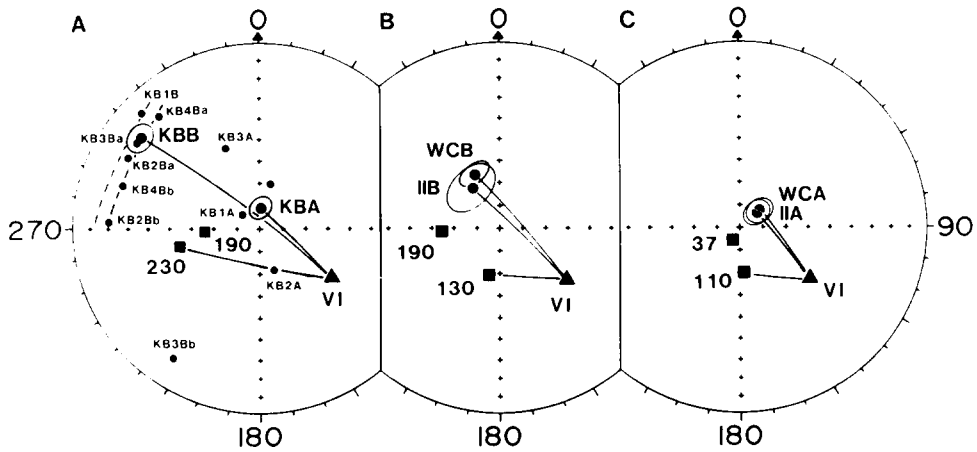


Fig. 8. Polar projections for the Northern Hemisphere showing: a) Karmutsen Basalts poles from Symons (1971a, KB1), Irving and Yole (1980, KB2), Schwarz *et al.* (1980, KB3) and Yole and Irving (1980, KB4) with their component means (KBA, KBB) and the small circle arc (parallel dashed lines) for the KBB poles; b) B component poles for the Westcoast Complex (WCB) and Island Intrusions (IIB); and c) A component poles for the Westcoast Complex (WCA) and Island Intrusions (IIA). The reference pole positions for cratonic North America (squares) with ages in Ma are from Irving and Irving (1982). Some poles are circumscribed by their cones or ovals of 95% confidence. The location of the sampling area is shown by a triangle.

Irving 1982), and indicates that the Karmutsen was extruded at an $18° \pm 6°$S latitude with a subsequent $\sim 39°$ of northward translation to their present position (Yole and Irving 1980). The arcuate distribution indicates minor ($\sim 20° \pm 13°$) relative block rotations about vertical axes between different sampling areas and provides a form of fold test (Morris 1981) that indicates the KBB component is primary. The weighted average KBB pole is located at 308°W, 22°N ($\alpha_{95} = 6°$) and indicates an overall $38 \pm 8°$ of clockwise rotation relative to the craton. Since neither of the WCA or WCB poles show the comparable low paleolatitudes or arcuate distribution, both are thus post-Upper Triassic and do not reflect the Late Paleozic origin of the Westcoast Complex.

Island Intrusions Poles

Symons (1971b) first reported a pole for the Jurassic Island Intrusions based on AF demagnetized data from 218 specimens from 24 sites distributed throughout Vancouver Island. Since then these specimens have been thermally demagnetized at 200, 450 and 560°C and analyzed using methods recently described by Symons (1983a, b) (Fig. 9). At 450°C specimens from 11 sites retain A component directions and from 10 sites

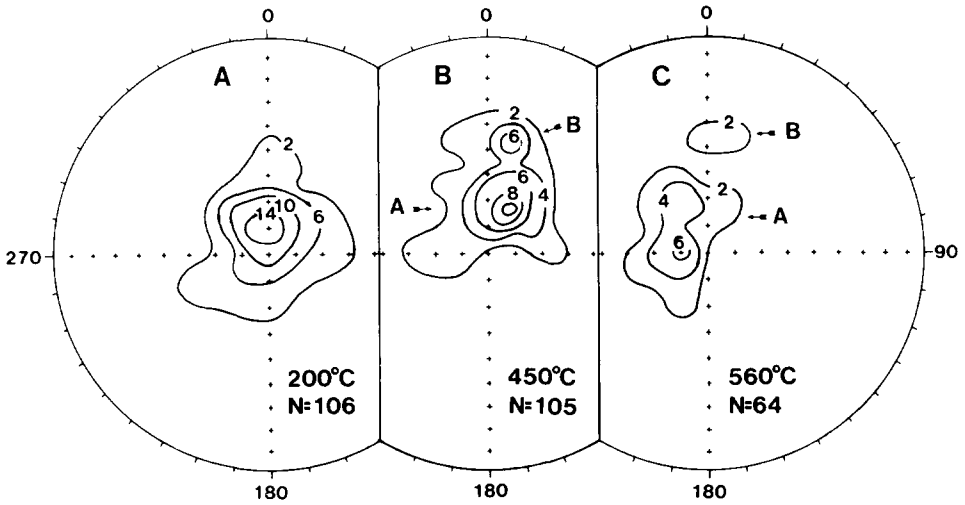


Fig. 9. Stereocountour plots for the residual directions of the Island Intrusions after thermal demagnetization at: a) 200°C; b) 450°C; and c) 560°C. Conventions as in Figure 2.

retain B component directions. Both A and B components survive at 560°C but in fewer sites so that either could be primary. Using the 450°C data, the IIA pole is at 59°W,79°N ($\delta_p = 8°$) (Table 2) and the IIB pole is at 325°W, 71°N ($\delta_p = 9°$, $\delta_m = 14°$). Systematic tectonic tilt on block faults may have occured on Vancouver Island (Muller 1977) causing these poles to deviate. Using the mapped attitudes at the three closest points within <8 km of each site for the Triassic and Jurassic strata, 56 attitudes were compiled which average to within 2° of horizontal, thereby indicating that tilt is not statistically significant. Comparing these poles to the Westcoast poles suggest that the IIA and WCA poles and the IIB and WCB poles are correlative.

DISCUSSION

Constraints

An unambiguous geotectonic synthesis for the Vancouver Island segment of the Wrangellia Subterrane is not yet possible from the paleomagnetic data. There are, however, some useful initial constraints on their interpretation.

The Wrangellia Subterrane originated as a late Paleozoic volcanic arc and eugeosynclinal assemblage (Monger and Price 1979), however, no paleomagnetic record from this time is preserved in the Westcoast Complex.

About 230 Ma ago at the time of rift extrusion of the Karmutsen Basalts, the subterrane was located at an $\sim 18^\circ \pm 6^\circ$S paleolatitude or $\sim 39^\circ \pm 6^\circ$ south of its present position relative to the North American craton when the primary KBB component poles originated. An $18^\circ \pm 6^\circ$N paleolatitude is unlikely because this would require an $\sim 180^\circ$ rotation of the subterrane. Since Upper Triassic time, the subterrane has been translated northward by $\sim 39^\circ \pm 6^\circ$ and rotated clockwise by $\sim 38^\circ \pm 8^\circ$ along with internal rotations between sampling areas of $\sim 20^\circ \pm 13^\circ$ about vertical axes.

Very limited paleomagnetic data from 13 sites in six small Paleocene or Eocene plutonic units from Vancouver Island (Symons 1971a) when averaged indicate negligible translation (2°S) and $\sim 15^\circ$ of clockwise rotation with large error limits.

The Eocene East Sooke Gabbro at the southernmost tip of Vancouver Island gives a pole position (Symons 1973a) that indicates $5^\circ \pm 7^\circ$ of northward translation combined with $20^\circ \pm 7^\circ$ of anticlockwise rotation. This is the only unit to show anticlockwise rotation and this feature is attributed to its location on the south side of the Leach River transcurrent fault zone (Muller 1977) which was apparently the locus for complex tectonic interactions with the Oregon Coast Ranges to the south. In the Oregon Coast Ranges, Simpson and Cox (1977) have found continuous clockwise rotation at a rate of $\sim 1^\circ$ per million years throughout the Cenozoic, and they have proposed two microplate tectonic models for these ranges to explain their results.

The Coast Plutonic Complex is a major tectonic welt on the east side of Terrane II. It was formed by the collision of Terrane II into the western margin of the North American craton during late Upper Jurassic to Eocene time (Monger *et al.* 1982). Paleomagnetic data is available from the Howe Sound, Bella Coola, Laredo Sound (excluding the Hawskbury Warp) and Prince Rupert areas along the length of the Coast Plutonic Complex (Symons 1969, 1973b, 1977a 1977b). This data is based on AF demagnetized specimens for the most part, but nevertheless they provide a relatively consistent pattern which is being substantiated by further thermal demagnetization work that is now in progress on some of the units. The Miocene and Eocene plutons yield concordant pole positions requiring no translation or rotation. As the ages of the plutons increase to ~ 80 Ma, ~ 110 Ma and ~ 130 Ma progressively, the amount of northward translation increases to $\sim 9^\circ$, 18° and 24°, and the amount of clockwise rotation to $\sim 46^\circ$, 39° and 48° respectively. Given that the Coast Plutonic Complex was formed by accretion of Terrane II, it is probably that Vancouver Island has a similar pattern of geotectonic motion. Beck *et al.* (1981) have discussed the significance of this far-sided and clockwise polar pattern for western North America.

Since Jones *et al.* (1977) originally defined the Wrangellian Subterrane, a considerable body of paleomagnetic data for Alaska has been accumulated which Stone *et al.* (1982) have summarized. Stone *et al.* (1982) show unequivocally that Wrangellia and the adjacent outboard terranes have moved systematically northward at a rate of ~ 6 cm a^{-1} or by $\sim 50°$ of latitude relative to the North American craton since Lower Cretaceous time. Their preferred but equivocal interpretation, because of large angular rotations, suggests southerly motion from Triassic to Early Jurassic time. The southerly motion is hard to reconcile with the previously discussed Karmutsen Basalts data.

Interpretation

Acceptance of an ~ 190 Ma age for the Westcoast Complex and Island Intrusions B components (WCB, IIB, Fig. 8b) to correspond to the age of migmatization and intrusion respectively implies $\sim 6° \pm 6°$ of northward translation and $\sim 20° \pm 8°$ of clockwise rotation. This creates two problems. First, it indicates that Vancouver Island moved northward by $\sim 33°$ between ~ 250 Ma and ~ 190 Ma and back southwared by about the same amount by ~ 130 Ma. This is both awkward and produces very high rates of translation. Second, and similarly, it implies $\sim 20°$ of clockwise rotation followed by $\sim 20°$ of counterclockwise rotation during the same two time intervals to fit with both the Karmutsen B component and Coast Plutonic Complex data. The most attractive alternative is to assume that the magnetization was acquired during a thermal event at ~ 130 Ma. This is attractive for three reasons. First, Terrane II impacted on the western margin of North America at this time to form major plutons along the westernmost side of the Coast Plutonic Complex. Second, heating at this time could be reflected in the updating of the K-Ar ages from the Island Intrusions to ~ 155 Ma. Third, the paleomagnetic data would then imply $\sim 23° \pm 7°$ of northward translation and $\sim 48° \pm 11°$ of clockwise rotation for Terrane II which is then reasonably consistent with the Coast Plutonic Complex data and the Karmutsen Basalts B component. An even younger age of remanence acquisition, down to ~ 110 Ma, is also possible. This simply increases the amount of northward translation by $\sim 10°$ and decreases the required clockwise rotation by $\sim 5°$. It does, however, result in a more awkward explanation for the A components.

There are two probably times of magnetization for the A components. One alternative is coincident with the K-Ar ages of Tertiary plutonic events found in their respective outcrop areas. In the Westcoast Complex there are minor quartz diorite stocks which give Lower Eocene ages in the $\sim 55 \pm 5$ Ma range, and in the Karmutsen Basalts and Island Intrusions

there are minor flows and plutons which give Upper Eocene ages in the $\sim 37 \pm 2$ Ma range. Using the reference poles for thes ages, the WCA, KBA and IIA poles indicate $2° \pm 7°$, $7° \pm 6°$, and $0° \pm 7°$ of northward translation, and $28° \pm 12°$, $20° \pm 9°$, and $26° \pm 13°$ of clockwise rotation respectively. Although possible, the high average clockwise rotation of $25° \pm 12°$ appears unlikely given the nonsignificant northward translation average of $3° \pm 7°$. The second alternative is to relate remanence acquisition to the ~ 110 Ma thermal event observed in the K-Ar dates for Vancouver Island (Muller 1977) which corresponds to the second major intrusive plutonic event found in the western Coast Plutonic Complex (Hutchison 1970). Under this hypothesis, the WCA, KBA and IIA poles indicate $15° \pm 7°$, $16° \pm 6°$ and $8° \pm 7°$ of northward translation, and $46° \pm 12°$, $42° \pm 9°$ and $47° \pm 13°$ of clockwise rotation respectively. The average values of $13° \pm 7°$ for northward translation and $45° \pm 12°$ for clockwise rotation then yield a pattern that is reasonably compatible with the other paleomagnetic data.

SUMMARY

The palemagnetic data for the Vancouver Island segment of Wrangellia can be most easily, but not uniquely explained, as follows. At the time of extrusion of the Karmutsen Basalts, their B component (KBB) indicates an $18° \pm 6°$S paleolatitude relative to the North American craton. From Upper Triassic to Upper Jurassic time, the subterrane moved northward from $\sim 39° \pm 6°$ to $\sim 23° \pm 7°$ south of its present location. This assumes that the WCB and IIA components were acquired at the time of initial collision of Terrane II with the craton to begin formation of the Coast Plutonic Complex. The Complex also indicates a similar $\sim 24°$ southern position for this time. From Upper Jurassic to mid-Cretaceous time, Terrane II and the Coast Plutonic Complex continued moving northward from $\sim 23° \pm 7°$ to $\sim 13° \pm 7°$ south of its present position. This assumes that the WCA, KBA and IIA components were acquired at $\sim$ 110 Ma or coincident with the thermal pulse found in the K-Ar age dates from dates from Vancouver Island and with the second major plutonic event found in the Coast Plutonic Complex. Plutons from the Complex of this age also indicate $\sim 18°$ of northward motion. They further indicate northward motion to a position of $\sim 9°$ south by ~ 80 Ma. By Eocene time, plutons both on Vancouver Island and in the Coast Plutonic Complex indicate that northward motion had essentially ceased.

The clockwise rotation of Terrane II and the Coast Plutonic Complex appears to be a relatively young event. From Upper Triassic to Upper Cretaceous time, all of the remanence components - KBB, WCB, IIB, KBA,

WCA, IIA and those from units in the Coast Plutonic Complex - all indicate clockwise rotations ranging between $38° \pm 8°$ and $48° \pm 11°$. In contrast, Eocene units in Terrane II and the Complex give concordant pole positions that suggest only a few degrees of clockwise rotation at most. This indicates that clockwise rotation occurred essentially between ~ 80 Ma and ~ 40 Ma as part of the final accretion process of Terrane II to the craton. The indicated rate and sense of rotation of $\sim 1°$ per million years corresponds to that found by Simpson and Cox (1977) in the Oregon Coast Ranges. Further similar northward displacements and clockwise rotations are observed for the same time frame in Terrane I on the east side of the Coast Plutonic Complex (Monger and Irving 1980, Symons 1983a, b).

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